Spectral Simulation Report

Automated Analysis

June 18, 2025

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1 Introduction

This report presents the results of spectral simulations derived from HDF5 dataset analysis. Each sample includes a spectral plot displayed in a full horizontal page and a detailed table summarizing the initial composition, actual positions, total pixels, and signal-to-noise ratio (SNR), along with simulation parameters such as temperature and electron density.

2 Sample: 10365

2.1 Simulation Parameters

Parameter	**Value**
Temperature (T)	7323 K
Electron Density (n_e)	5.86e+14 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	34.75

Table 1: Simulation parameters and additional metrics for sample 10365.

2.2 Composition Analysis

Element	**Initial Composition (%)**	**Actual Positions**
Al	0.00	0.0
Ar	0.84	657.0
C	11.91	226.0
Ca	5.45	150.0
Cl	15.37	12.0
Co	11.62	1402.0
Cr	2.75	397.0
Fe	4.98	2696.0
Mg	13.72	316.0
Mn	0.00	0.0
N	0.00	0.0
Na	9.11	322.0
Ni	12.28	478.0
O	0.00	0.0
S	3.79	30.0
Si	8.17	220.0
Ti	0.00	0.0
background	N/A	698.0
Total Positions		7604.0

Table 2: Composition analysis for sample 10365.

2.3 Spectral Plot

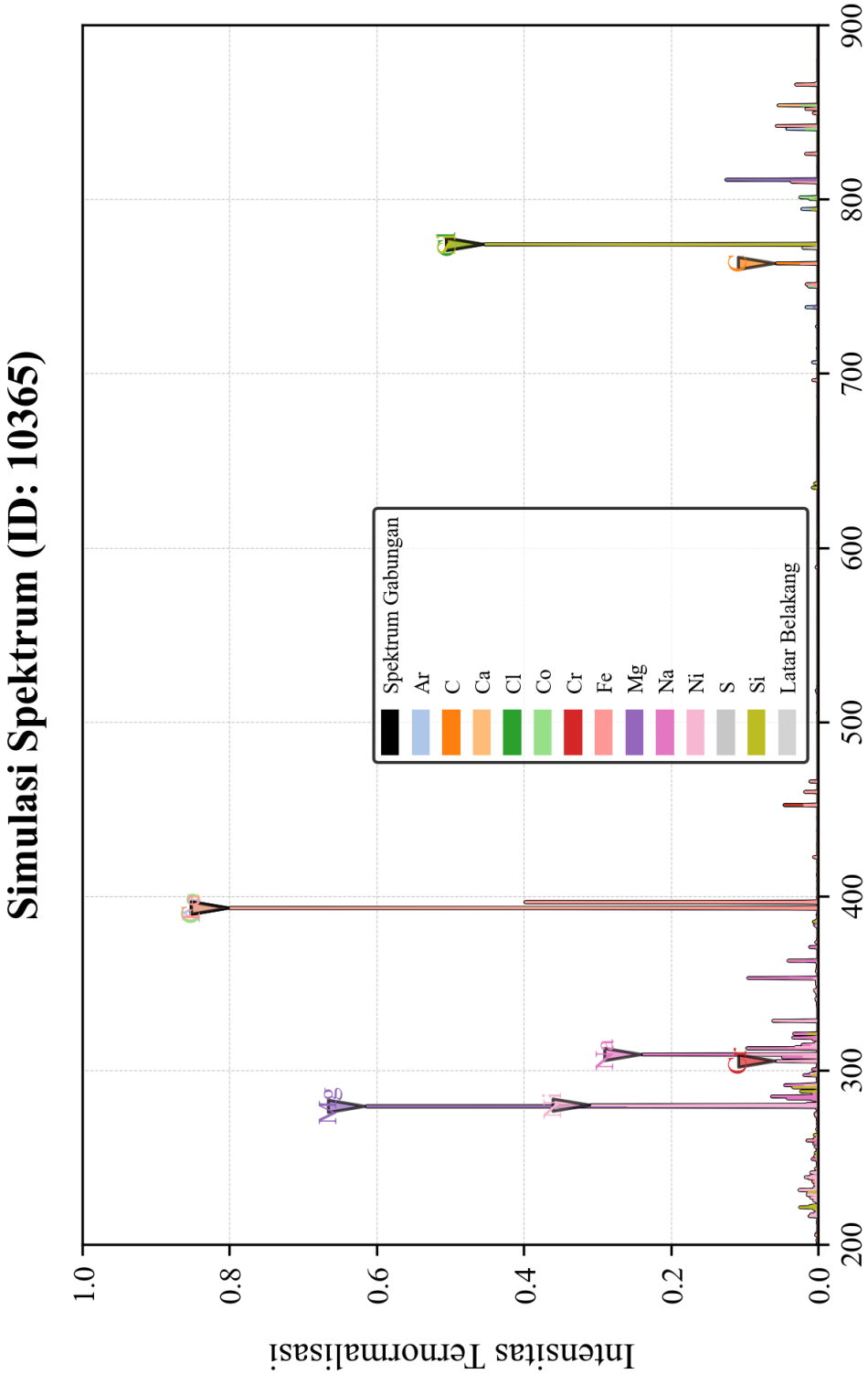

Figure 1: Spectral simulation plot for sample 10365 in full horizontal layout.

3 Sample: 13810

3.1 Simulation Parameters

Parameter	Value
Temperature (T)	8636 K
Electron Density (n_e)	3.27e+17 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	31.14

Table 3: Simulation parameters and additional metrics for sample 13810.

3.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	15.00	425.0
Ar	0.00	0.0
C	0.00	0.0
Ca	5.95	150.0
Cl	11.02	12.0
Co	0.00	0.0
Cr	9.42	397.0
Fe	0.00	0.0
Mg	6.49	341.0
Mn	1.22	706.0
N	0.00	0.0
Na	14.91	330.0
Ni	8.67	478.0
O	16.69	168.0
S	0.85	25.0
Si	9.77	218.0
Ti	0.00	0.0
background	N/A	1932.0
Total Positions		5182.0

Table 4: Composition analysis for sample 13810.

3.3 Spectral Plot

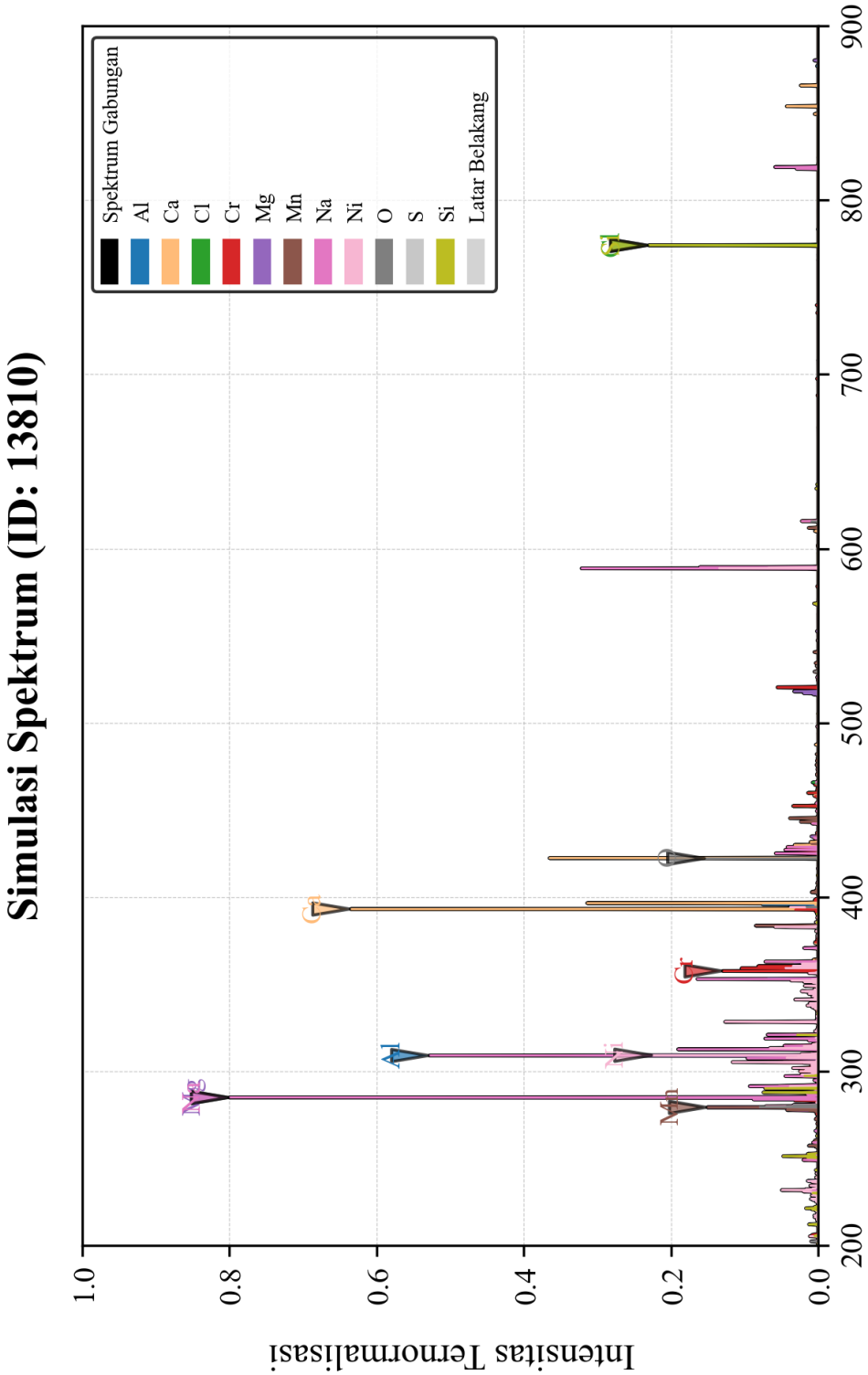

Figure 2: Spectral simulation plot for sample 13810 in full horizontal layout.

4 Sample: 193

4.1 Simulation Parameters

Parameter	Value
Temperature (T)	5909 K
Electron Density (n_e)	5.34e+14 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	37.52

Table 5: Simulation parameters and additional metrics for sample 193.

4.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	4.88	140.0
Ar	8.57	355.0
C	9.52	167.0
Ca	4.64	150.0
Cl	6.81	12.0
Co	3.67	862.0
Cr	6.37	397.0
Fe	0.00	0.0
Mg	3.93	160.0
Mn	9.55	673.0
N	4.90	396.0
Na	5.52	322.0
Ni	0.30	478.0
O	2.06	8.0
S	9.80	25.0
Si	11.34	205.0
Ti	8.16	1181.0
background	N/A	1210.0
Total Positions		6741.0

Table 6: Composition analysis for sample 193.

4.3 Spectral Plot

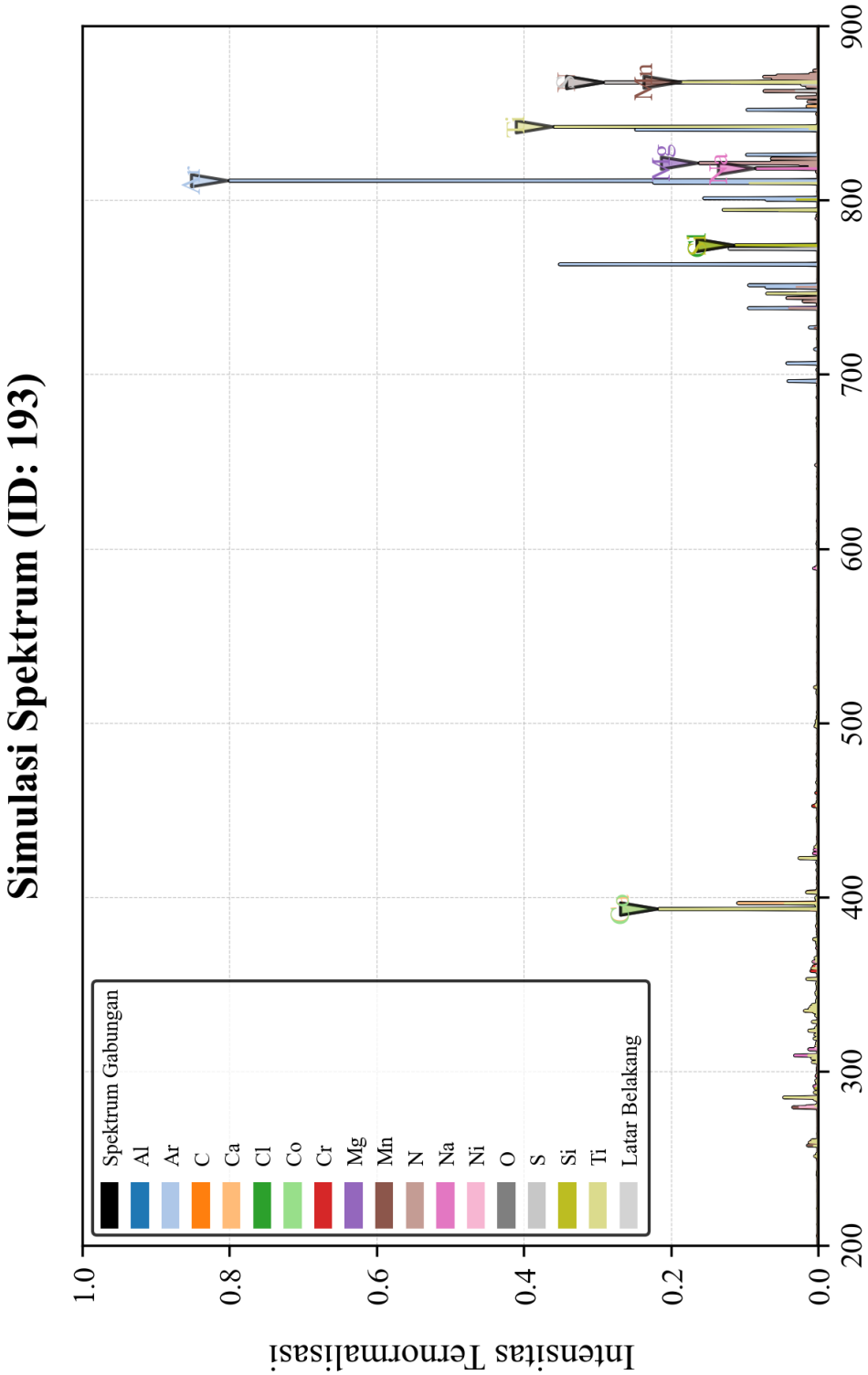

Figure 3: Spectral simulation plot for sample 193 in full horizontal layout.

5 Sample: 674

5.1 Simulation Parameters

Parameter	Value
Temperature (T)	12576 K
Electron Density (n_e)	2.60e+15 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	41.82

Table 7: Simulation parameters and additional metrics for sample 674.

5.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	6.23	462.0
Ar	1.46	727.0
C	11.79	688.0
Ca	16.90	150.0
Cl	7.05	35.0
Co	0.00	0.0
Cr	7.85	397.0
Fe	0.00	0.0
Mg	7.85	352.0
Mn	2.19	706.0
N	0.00	0.0
Na	6.67	322.0
Ni	10.93	478.0
O	3.82	255.0
S	4.90	226.0
Si	0.85	219.0
Ti	11.51	1189.0
background	N/A	1053.0
Total Positions		7259.0

Table 8: Composition analysis for sample 674.

5.3 Spectral Plot

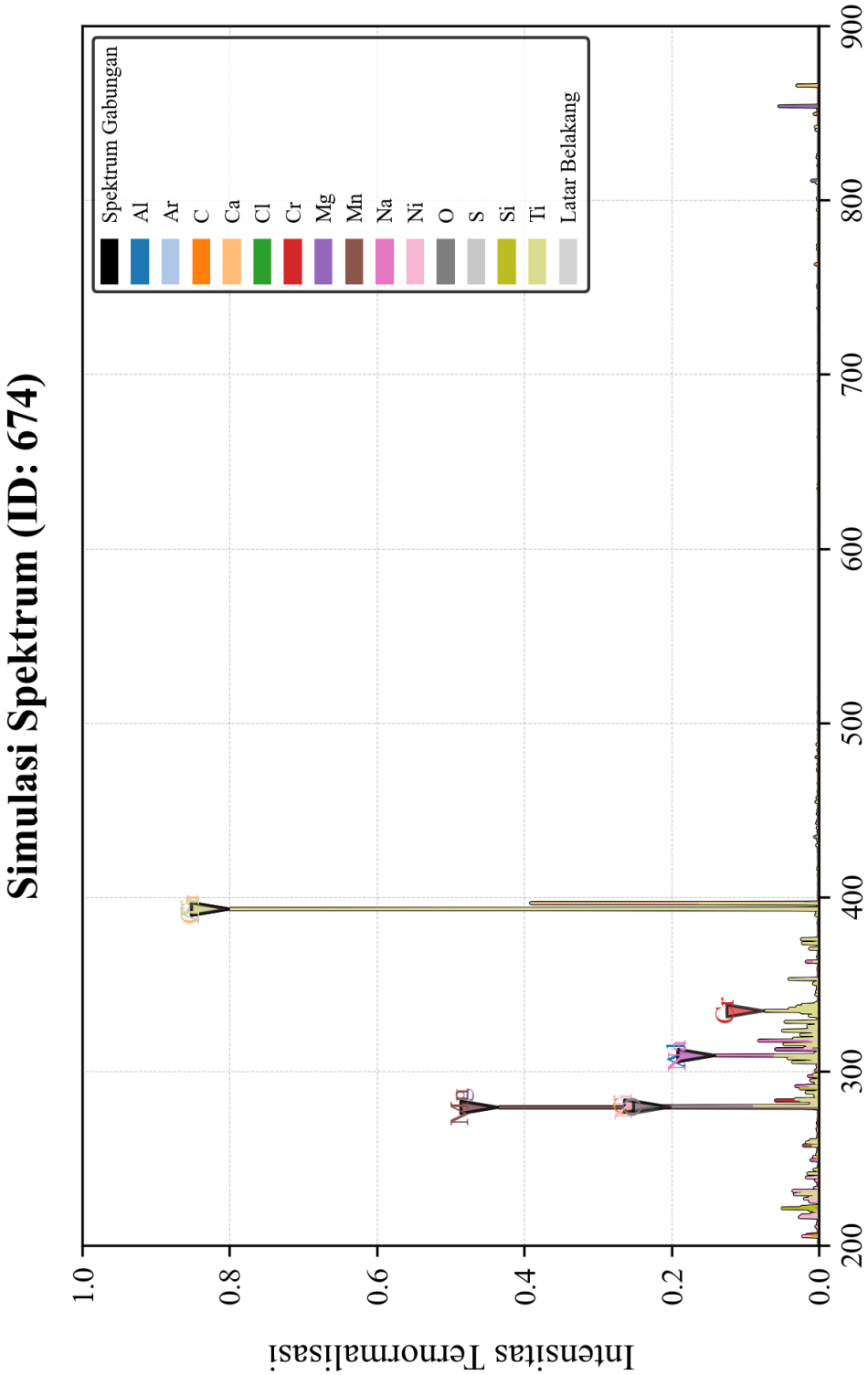

Figure 4: Spectral simulation plot for sample 674 in full horizontal layout.

6 Sample: 1636

6.1 Simulation Parameters

Parameter	Value
Temperature (T)	8737 K
Electron Density (n_e)	2.48e+17 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	32.07

Table 9: Simulation parameters and additional metrics for sample 1636.

6.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	9.88	424.0
Ar	0.78	629.0
C	3.36	222.0
Ca	10.11	150.0
Cl	3.55	12.0
Co	0.52	1371.0
Cr	2.84	397.0
Fe	0.00	0.0
Mg	10.17	346.0
Mn	0.87	706.0
N	7.93	396.0
Na	11.72	330.0
Ni	4.95	478.0
O	14.61	170.0
S	7.11	68.0
Si	7.37	218.0
Ti	4.23	1183.0
background	N/A	797.0
Total Positions		7897.0

Table 10: Composition analysis for sample 1636.

6.3 Spectral Plot

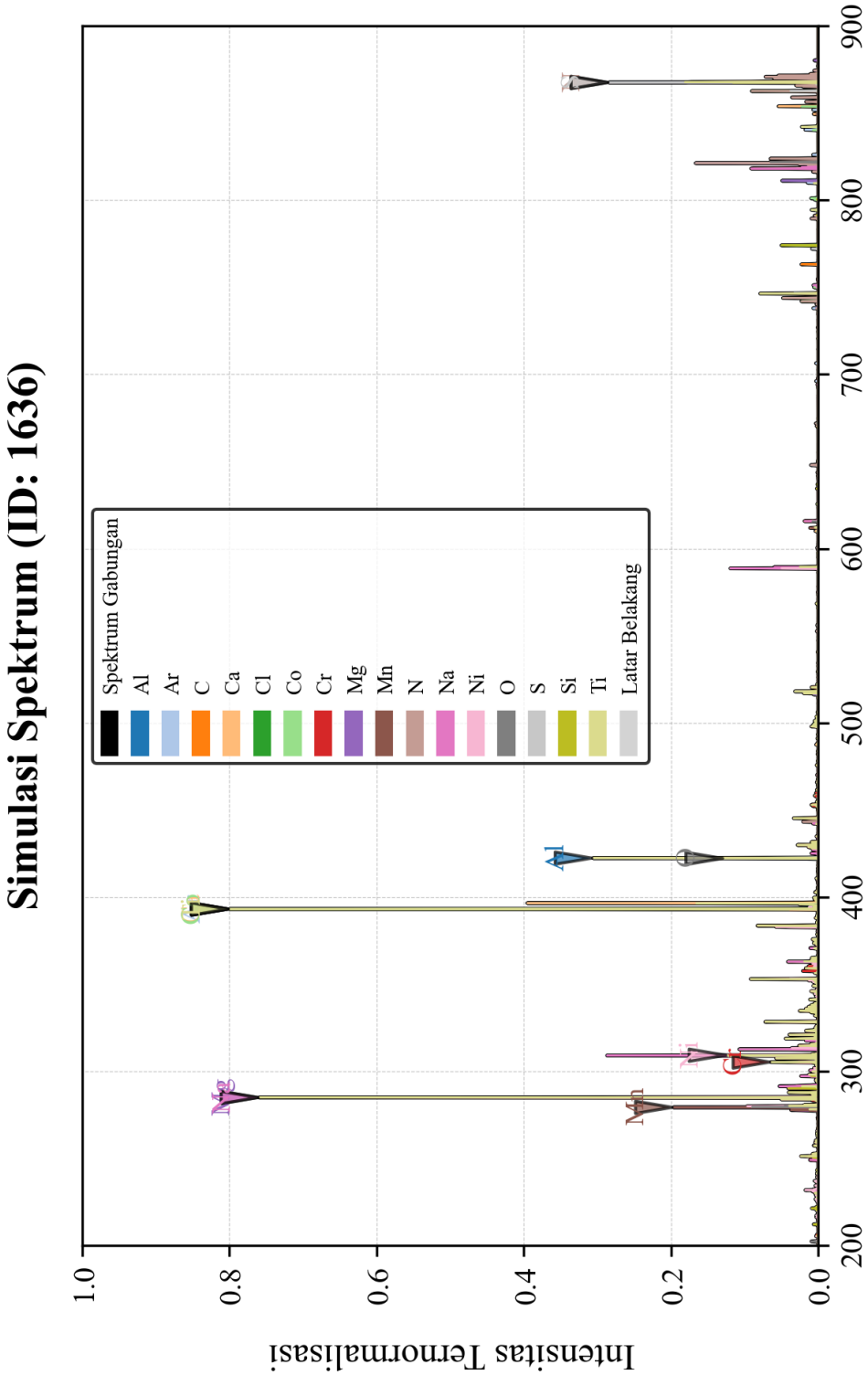

Figure 5: Spectral simulation plot for sample 1636 in full horizontal layout.